The Impact of the WTO on Bilateral Trade

Structural gravity estimates with PPML, 2000-2018

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2026-01-01

1 Introduction

Does WTO membership raise trade between members? Cross-sectional gravity estimates usually find large positive effects, while Rose (2004) found little evidence once country-pair heterogeneity is taken seriously. We revisit the question on annual data for 2000-2018 using a structural gravity model estimated by Poisson pseudo-maximum likelihood (PPML, Santos Silva and Tenreyro, 2006), with exporter-year, importer-year and country-pair fixed effects (Yotov et al., 2016). We also ask whether the effect differs with the development level of the two partners.

This project was carried out for the Quantitative International Economics course of the M2 at Université Paris Dauphine-PSL. The R code behind every table and figure is in the source file `wto_gravity_ppml.qmd`.

2 Data

We combine three sources:

- **USITC Dynamic Gravity Dataset** (release 2.1): WTO and GATT membership, trade agreements, distance, common language, contiguity, colonial ties and GDP, by country pair and year.
- **USITC International Trade and Production Database for Estimation** (ITPDE, release 3): bilateral trade flows, including zero flows and domestic flows (a country's sales to itself).
- **Gravity_M2IT**, a gravity database provided in the course: World Bank income group of each country.

We keep observations with non-missing trade flows, GDP and distance. The resulting sample covers 2000-2018.

2.1 Development level and pair types

Countries are grouped into three levels from the World Bank income classification: **High (H)** for high income, **Intermediate (I)** for upper and lower middle income, and **Low (L)** for low income. Each pair is then assigned one of six symmetric types, independent of the direction of the flow: HH, HI, HL, II, LI and LL.

Table 1: Estimation sample

Characteristic	Value
Total observations	658,966
Number of countries	198
Number of unique pairs	36,200
Number of years	19
Time period	2000 - 2018
International trade observations	656,132
Domestic trade observations	2,834
Share of zero trade flows	25.6%
Number of switcher pairs	6,502

Table 2: Sample by year

Year	Observations	Countries	Total trade	% zero flows	% both WTO
2000	32,607	186	42,637,877	32.9	56.8
2001	32,927	187	42,004,671	31.8	58.8
2002	34,316	192	43,509,024	31.9	56.4
2003	34,299	192	50,337,187	30.7	58.0
2004	34,326	192	58,742,679	29.6	59.6
2005	34,336	192	63,215,626	28.8	60.5
2006	35,079	194	69,711,647	27.5	59.2
2007	35,345	195	78,870,846	26.3	60.3
2008	35,363	195	86,077,472	25.5	61.9
2009	35,359	195	77,299,361	25.1	61.9
2010	35,955	197	85,158,318	24.3	61.6
2011	35,361	195	98,246,646	23.7	61.9
2012	35,510	196	99,638,794	23.3	64.7
2013	35,503	196	102,669,657	22.4	66.3
2014	35,164	195	103,458,665	22.8	67.8
2015	34,806	194	95,743,214	21.7	69.4
2016	34,632	193	84,296,583	20.5	71.4
2017	34,633	193	87,923,900	20.1	71.4
2018	33,445	189	94,009,242	19.3	73.9

Total trade in millions of current US dollars (ITPDE).

Table 3: Descriptive statistics by pair type

Pair type	Obs	Mean trade	Share exporter in WTO	Share both in WTO	Mean distance (km)
HH	80,193	13,536.1	0.85	0.71	6,930
HI	244,572	442.5	0.81	0.64	8,182
HL	56,203	18.3	0.83	0.69	7,228
II	184,105	1,448.1	0.76	0.58	8,550
LI	84,136	13.1	0.78	0.61	7,374
LL	9,757	113.4	0.79	0.64	3,923

2.2 Descriptive statistics

2.3 WTO accessions during the sample period

Identification in the structural model comes from pairs whose joint membership changes over the period. Table 4 lists the countries that joined the WTO between 2000 and 2018; most are intermediate-income economies.

Table 4: Countries joining the WTO during the sample period

Country	Accession year	Development level
CHN	2001	I
LTU	2001	H
MDA	2001	I
ARM	2003	I
MKD	2003	I
KHM	2004	I
NPL	2004	I
SAU	2005	H
TON	2007	I
VNM	2007	I
CPV	2008	I
UKR	2008	I
MNE	2012	I
RUS	2012	I
VUT	2012	I
WSM	2012	I
LAO	2013	I
TJK	2013	L
YEM	2014	L
KAZ	2015	I
SYC	2015	H
AFG	2016	L
LBR	2016	L

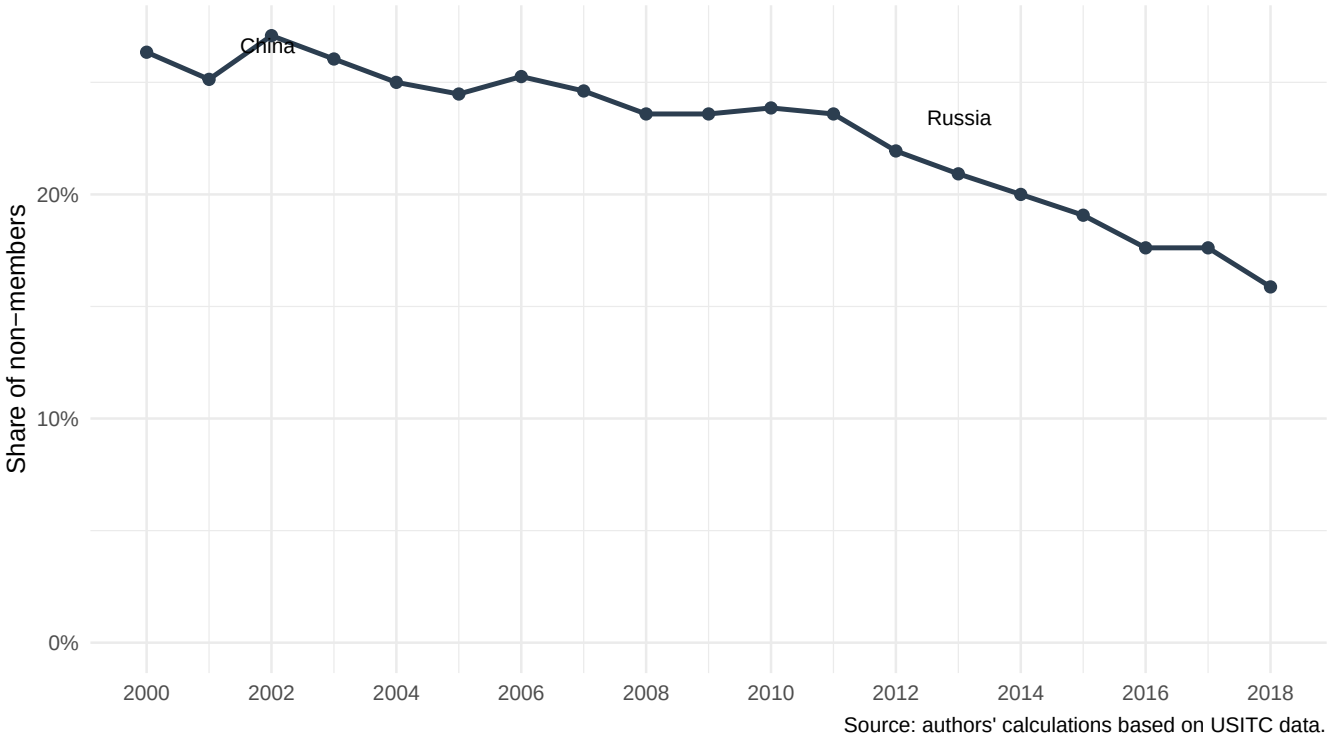


Figure 1: Share of countries outside the WTO, estimation sample

3 Empirical strategy

We estimate four specifications by PPML, which handles zero flows and is consistent under heteroskedasticity:

$$X_{ijt} = \exp\left(\beta\,\mathrm{WTO}_{ijt} + \gamma\,\mathrm{FTA}_{ijt} + \sum_{s}\delta_s\,\mathrm{INTL}_{ij}\times\mathbb{1}[t=s] + \pi_{it} + \chi_{jt} + \mu_{ij}\right)\times\varepsilon_{ijt}$$

where X_{ijt} is trade from i to j in year t, WTO_{ijt} equals one when both countries are WTO members, and FTA_{ijt} indicates a free trade agreement. The exporter-year and importer-year fixed effects π_{it} and χ_{jt} absorb multilateral resistance and country size. The interactions of the international border dummy with year dummies (2000, 2004, 2009 and 2014) control for the common decline in the cost of international relative to domestic trade.

- **(1) Traditional**: no pair fixed effects; distance, common language, contiguity and colonial ties enter instead.
- **(2) Structural**: pair fixed effects μ_{ij} absorb all time-invariant pair characteristics, so β is identified only from pairs whose joint membership changes during the period.
- **(3) and (4)**: same as (1) and (2), with WTO_{ijt} interacted with the six pair types.

A dummy for exactly one partner being a member cannot be included: with exporter-year and importer-year fixed effects it is an exact linear combination of WTO_{ijt} and the fixed effects. Standard errors are clustered by exporter and by importer.

4 Results

Table 5: WTO membership and bilateral trade, PPML estimates (2000-2018)

	Average effect		By pair type	
	(1) Trad.	(2) Struct.	(3) Het. trad.	(4) Het. struct.
Both WTO members	1.917***	−0.197*		
	(0.276)	(0.103)		
FTA	−0.939***	0.172***	−1.038***	0.174***
	(0.167)	(0.062)	(0.167)	(0.062)
Log distance	−1.608***		−1.503***	
	(0.126)		(0.120)	
Common language	1.004***		0.990***	
	(0.207)		(0.194)	
Contiguity	−0.783***		−0.697***	
	(0.164)		(0.146)	
Colonial tie	−0.351*		−0.470*	
	(0.202)		(0.248)	
Both WTO x HH			2.398***	−0.373***
			(0.259)	(0.073)
Both WTO x HI			1.649***	−0.124
			(0.283)	(0.098)
Both WTO x HL			2.150***	−0.427**
			(0.331)	(0.218)
Both WTO x II			2.024***	0.080
			(0.432)	(0.091)
Both WTO x LI			2.425***	−0.111
			(0.380)	(0.175)
Both WTO x LL			5.847***	0.428
			(0.609)	(0.568)
Num.Obs.	658 966	646 149	658 966	646 149
Pair FE	No	Yes	No	Yes
Exporter-year FE	Yes	Yes	Yes	Yes
Importer-year FE	Yes	Yes	Yes	Yes
Border x year	Yes	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Standard errors clustered by exporter and importer in parentheses. Structural specifications drop pairs with only zero flows.

Traditional versus structural estimates. Without pair fixed effects, joint WTO membership is associated with 580% more trade (column 1). This estimate mostly reflects differences across pairs: members trade more with each other for many reasons unrelated to membership. Once pair fixed effects are included, the coefficient becomes -0.197 ($p = 0.06$), i.e. -17.9% in percentage terms, and is only significant at the 10% level (column 2). Within the 2000-2018 accessions, we find no evidence that joining the WTO raised trade with other members. Free trade agreements, by contrast, raise trade by about 19% in the structural model.

Heterogeneity by development level. The structural estimates by pair type (column 4) are negative and significant for pairs involving a high-income country (HH and HL), and not significantly different from zero for the other pair types. The large positive coefficients of column 3 again disappear once pair fixed effects are included.

Robustness. Table 6 replaces the four border-year interactions with a full set of border-year dummies. The coefficients are close to the baseline ones.

Table 6: Robustness to the globalization controls

	(2) Struct.	(2') All border-year	(4) Het. struct.	(4') All border-year
Both WTO members	−0.197*	−0.205*		
	(0.103)	(0.112)		
FTA	0.172***	0.143***	0.174***	0.145***
	(0.062)	(0.050)	(0.062)	(0.049)
Both WTO x HH			−0.373***	−0.404***
			(0.073)	(0.083)
Both WTO x HI			−0.124	−0.133
			(0.098)	(0.108)
Both WTO x HL			−0.427**	−0.447**
			(0.218)	(0.218)
Both WTO x II			0.080	0.065
			(0.091)	(0.091)
Both WTO x LI			−0.111	−0.131
			(0.175)	(0.177)
Both WTO x LL			0.428	0.527
			(0.568)	(0.594)
Num.Obs.	646 149	646 149	646 149	646 149

* p < 0.1, ** p < 0.05, *** p < 0.01

All columns include pair, exporter-year and importer-year fixed effects. Columns (2') and (4') interact the border dummy with every year (reference 2000).

5 Limitations

The structural estimate rests on the countries that joined during the period, mostly intermediate-income economies whose accession commitments were negotiated over long periods, so part of the trade response may predate formal membership and be absorbed by the fixed effects. Domestic flows are available for only 2,834 country-year observations, which limits how precisely the border-year controls capture globalization. Larch et al. (2019) show that adding domestic flows for all countries and allowing for non-discriminatory effects of membership can change the conclusions substantially.

6 References

- Larch, M., Monteiro, J.-A., Piermartini, R. and Yotov, Y. V. (2019). On the effects of GATT/WTO membership on trade: They are positive and large after all. *WTO Staff Working Paper* ERSD-2019-09.
- Rose, A. K. (2004). Do we really know that the WTO increases trade? *American Economic Review*, 94(1), 98-114.
- Santos Silva, J. M. C. and Tenreyro, S. (2006). The log of gravity. *Review of Economics and Statistics*, 88(4), 641-658.
- Yotov, Y. V., Piermartini, R., Monteiro, J.-A. and Larch, M. (2016). *An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model*. WTO and UNCTAD.